

Table 1. Accuracy and runtime of barcode clustering across three benchmarks.

Panels A and B: 10,000 true barcodes and approximately 1 million reads per library. Panel C: 100,000 true barcodes and 24,996,128 reads.

Method	FN ↓	FP ↓	F1 (%) ↑	Read accuracy (%) ↑	Runtime ^b (s) ↓
A Random N20: substitutions + indels (60 libraries)					
barbac Hamming	124.92	18.30	99.28004 ± 0.0758	99.997077	1.40
barbac LV + Poisson	124.90	18.25	99.28038 ± 0.0752	99.997083	1.76
Shepherd ^a	126.53	19.83	99.26421 ± 0.0766	99.996495	4.67
Starcode sphere	129.72	22.32	99.23569 ± 0.0797	99.982520	3.29
Starcode MP	118.18	142.98	98.69582 ± 0.1112	99.983934	3.22
Bartender	125.17	48.03	99.13063 ± 0.0921	99.930972	1.89
B Anchored N20 + AA/TT: substitutions + indels (60 libraries)					
barbac Hamming	127.70	24.90	99.23302 ± 0.0736	99.993137	1.52
barbac LV + Poisson	127.67	24.08	99.23726 ± 0.0741	99.994257	2.05
Shepherd ^a	129.65	25.05	99.22239 ± 0.0796	99.993818	13.45
Starcode sphere	132.63	27.65	99.19431 ± 0.0829	99.986583	3.77
Starcode MP	119.95	191.83	98.44674 ± 0.1384	99.979174	3.67
Bartender	127.73	137.97	98.67218 ± 0.0786	99.906649	2.31
C Milo / Johnson published simulation (one fixed reference)					
barbac Hamming	469	87	99.72147	99.999088	17.50
barbac LV + Poisson	471	83	99.72246	99.999476	33.65
Shepherd ^a	470	85	99.72196	99.999132	165.23
Starcode sphere	771	353	99.43682	99.847880	170.68
Starcode MP	532	560	99.45408	99.981789	193.39
Bartender	496	795	99.35546	99.935342	41.77

Panels A and B: FN, FP and read accuracy are means; F1 is mean ± standard deviation; runtime is median. Panel C reports one observation. Bold and pale orange shading identify the best observed value for each metric within each panel.

FN, false negatives: true identities absent from inferred centroids. FP, false positives: inferred identities absent from truth. Exact-centroid F1 = 200TP/(2TP + FN + FP), including zero-read true identities. Read accuracy is the percentage of all reads assigned to their true barcode; incorrect and unassigned reads both count as errors.

^a Shepherd uses distance 3, Bayes-factor threshold -4, and the generating substitution rate 0.004 in panels A/B; panel C uses automatic rate estimation. This is the completed post hoc sensitivity configuration; all other methods retain their original settings and results.

^b All workflows use one thread. Runtime includes fresh-worker/tool startup, required input conversion and centroid/member export; common staging and scoring are excluded. Shepherd was measured in a later session, so its timing comparisons are descriptive.

The accompanying Table 1 methods provide simulation details, complete settings, paired statistical comparisons and measurement provenance.